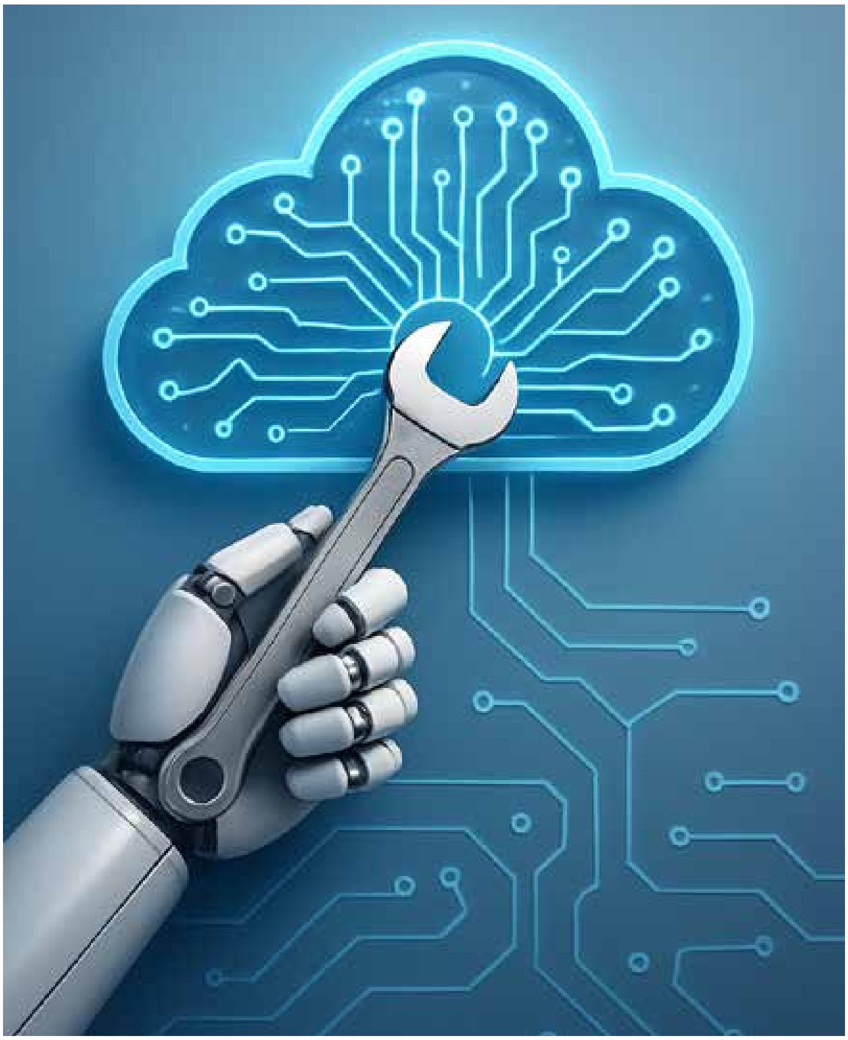
When AI Meets DevOps to Build Self-Healing Systems

Traditional DevOps, with its rule-based automation, is struggling to work effectively in today's complex tech world. But when combined with AIOps it can lead to IT systems that predict failures and solve issues without human intervention.

In the fast-paced and ever-changing world of software development and IT operations, automation is a great asset. From CI/CD pipelines to provisioning infrastructure, DevOps has equipped teams to construct and deploy software faster than ever. But as systems become more complex, distributed, and data-rich, automation in isolation is not enough.

This is where artificial intelligence for IT operations (AIOps) enters the conversation. By embedding AI and machine learning with DevOps practices, AIOps shifts the paradigms beyond a workflow of defined rules. Not only does AIOps analyse data patterns and detect anomalies, it can also anticipate failures and take preemptive action with little or no human assistance.



Why automation alone is no longer enough

DevOps has relied on automation for a long time to automate repetitive tasks like continuous integration, testing, deployment, and managing infrastructure. The benefit of automation is that it can eliminate repetitive manual work, speed up the delivery cycle, and reduce human errors. However, traditional forms of DevOps automation are rule based (such as 'if x happens, do y').

This is where the limit of automation occurs.

Modern IT ecosystems are dynamic and complex, such as cloud-native applications, microservices, containers, and systems distributed across the globe. With thousands of events and logs created every second, unplanned and unexpected issues do not always follow a pattern. Rule-based automation cannot adapt to new scenarios, unknowns, or subtle signals embedded in data in real-time. In such environments, we don't need just a fast response, but a smart one. This is why businesses are now turning to AI-powered solutions that learn, adapt, and evolve beyond traditional scripts.

Defining 'self-healing systems' in modern IT

A self-healing system is an intelligent system that recognises when it has an operations problem, figures out why it is having it, and can often fix it before